

$$X \downarrow \text{id}_X$$

$$\text{curly arrow } X \text{ (downward)}$$

$$\text{ev}_X: X^* \otimes X \rightarrow \mathbf{1}$$

$$\text{curly arrow } X \text{ (downward)}$$

$$\text{ev}_{X^*} \circ (\delta_X \otimes 1_{X^*}): X \otimes X^* \rightarrow \mathbf{1}$$

$$X \uparrow \text{id}_{X^*}$$

$$\text{curly arrow } X \text{ (upward)}$$

$$\text{coev}_X: \mathbf{1} \rightarrow X \otimes X^*$$

$$\text{curly arrow } X \text{ (upward)}$$

$$(\text{id} \otimes \delta_X^{-1}) \text{coev}_{X^*}: \mathbf{1} \rightarrow X^* \otimes X$$